## Assignment

### SECTION-A

**Multiple Choice Questions (MCQ) Type and Assertion – Reason Based Questions**

1. Position $x$ of a particle changes with time $t$ as $x = 3t^2 + 2t + 9$. Displacement of particle during time interval between $t$ = 2 s and $t$ = 5 s is ($x$ is in metre)
   (1) 69 m (2) 59 m
   (3) 71 m (4) 72 m

2. A car moving along straight line in single direction moves 150 m from station $A$ to station $B$ in 10 s and 100 m from station $B$ to station $C$ in same direction in 15 s without stopping in between at $B$. The average velocity of car when it moves from $A$ to station $C$ as described above will be
   (1) 15 m/s (2) 6 m/s
   (3) 10 m/s (4) 8 m/s

3. The position ($x$) of a body moving along a straight line at time $t$ is given by $x = 3t^2 - 5t + 2$. Then its velocity at $t$ = 2 s is ($x$ is in meter)
   (1) 7 m/s (2) 5 m/s
   (3) 12 m/s (4) 6 m/s

4. The displacement of a particle starting from rest is given by $x = 15t^2 - 2t^3$. The time (in s) at which the particle comes to rest again is ($x$ is in metres and $t$ is in seconds)
   (1) 5 (2) 4
   (3) 3 (4) 6

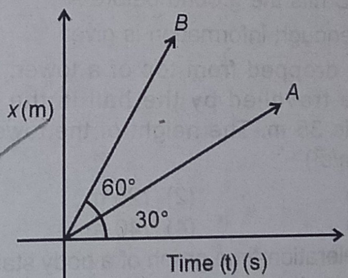
5. Velocity of a particle moving along $x$-axis varies with time $t$ as $v = (5t^2 - 7)$ m/s where $t$ is in seconds. The average acceleration of the particle from $t$ = 2 s to $t$ = 4 s is
   (1) 20 m/s$^2$ (2) 40 m/s$^2$
   (3) 30 m/s$^2$ (4) 50 m/s$^2$

6. A car starts from rest with its acceleration $a$ (in m/s$^2$) varying with time $t$ (in s) as $a = 3t + 4$. The velocity of car at $t$ = 6 seconds will be
   (1) 120 m/s (2) 78 m/s
   (3) 130 m/s (4) 94 m/s

7. A car starts moving from rest with a constant acceleration of 5 m/s$^2$ along a straight line. The distance travelled by it in the first 2 seconds is
   (1) 20 m (2) 10 m
   (3) 5 m (4) 40 m

8. A car was moving at a rate of 18 kmph when the brakes were applied. If it comes to rest after travelling a distance of 5 m then its average retardation is
   (1) 5 m/s$^2$ (2) 3 m/s$^2$
   (3) 2.5 m/s$^2$ (4) 7.5 m/s$^2$

9. A body is thrown vertically upward with a speed 20 m/s. Under the effect of gravity ($g$ = 10 m/s$^2$) what is the distance travelled by the body in first second of its motion?
   (1) 20 m (2) 15 m
   (3) 25 m (4) 10 m

10. 


The position-time graph of two objects $A$ and $B$ is shown in the figure. What is the ratio of velocity of object $A$ with the velocity of object $B$?
   (1) 1 : 3 (2) 3 : 2
   (3) 3 : 1 (4) $\sqrt{3}$ : 2

11. Which of the following is correct statement?
    (1) Magnitude of average velocity is always equal to the magnitude of average speed.
    (2) Magnitude of instantaneous velocity is always equal to the magnitude of instantaneous speed.
    (3) Magnitude of average acceleration and instantaneous acceleration is always same
    (4) In a uni-directional motion displacement always has a magnitude greater than distance travelled.

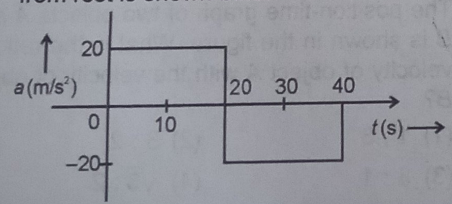
12. The acceleration $a$ in $m/s^2$ of a particle in $x$-direction is given by $a = 3t^2 + 2t + 2$, where $t$ is the time in seconds. If the particle starts with a velocity of 2 m/s, at $t = 0$, from a position $x = 2$ m along $x$-axis then position of the particle at the end of 2 seconds is
    (1) 18 m (2) 14 m
    (3) $\frac{50}{3}$ m (4) $\frac{39}{3}$ m

13. A particle starts at $t = 0$ at $x = 3$ m from rest. It moves with constant acceleration of 4 $m/s^2$ along $x$-axis. The position of particle at the end of $t = 3$ s is
    (1) 21 m (2) 15 m
    (3) 18 m (4) 24 m

14. Ball $A$ is thrown vertically upward with a speed of 19.6 m/s from the top edge of a high building. As it passes the edge on the way down, a second ball $B$, is thrown downward at 19.6 m/s. Which of the following is correct?
    (1) Ball $A$ hits the ground before $B$
    (2) The two balls hit the ground at the same time
    (3) Ball $B$ hits the ground before $A$
    (4) Not enough information is given

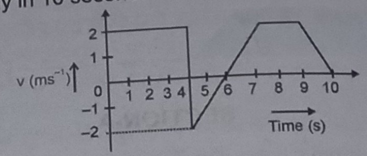
15. A ball is dropped from top of a tower. The distance travelled by the ball in the last second is 35 m. The height of the tower is ($g = 10$ $m/s^2$)
    (1) 80 m (2) 125 m
    (3) 95 m (4) 140 m

16. The acceleration-time graph of a body starting from rest is shown below




The average acceleration of the body from $t = 0$ s to $t = 40$ s is
    (1) 20 $m/s^2$ (2) 10 $m/s^2$
    (3) 0 $m/s^2$ (4) 15 $m/s^2$

17. The velocity-time graph of a body moving in a straight line is shown below, find the displacement and the distance travelled by the body in 10 second.




    (1) 12 m, 12 m (2) 16 m, 18 m
    (3) 16 m, 16 m (4) 12 m, 16 m

18. A train of 150 m length is going towards north direction at a speed of 10 m/s. A bird flies at a speed of 5 m/s towards the south direction parallel to the railway track. Find the time taken by the bird to cross the train.
    (1) 30 s (2) 12 s
    (3) 15 s (4) 10 s

## Assertion – Reason Based Questions

**Directions:** In question numbers 19 and 20, two statements are given one labelled **Assertion (A)** and the other labelled **Reason (R)**. Select the correct answer from the following options.

(1) Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of the Assertion (A)
(2) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of the Assertion (A)
(3) Assertion (A) is true, but Reason (R) is false
(4) Assertion (A) is false, but Reason (R) is true

19. **Assertion (A):** If velocity at an instant is zero then acceleration at that instant will also be zero.
    **Reason (R):** Area under velocity-time graph gives average acceleration during a time-period.

20. **Assertion (A):** For a body in motion displacement may be zero even if distance travelled is non-zero during a span of time.
    **Reason (R):** Displacement is a vector quantity while distance travelled is a scalar quantity.

## SECTION-B

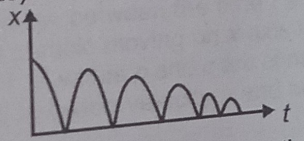
### Very Short Answer (VSA) Type Questions

21. A person travels along a straight road for the first half of the distance with a speed $v_1$ and the second half of the distance with speed $v_2$. What is the average speed of the person?
22. What is the acceleration of a body when its velocity-time graph is (i) perpendicular to time axis (ii) parallel to time axis?
23. A jet airplane travelling at the speed of 500 km $h^{-1}$ ejects its products of combustion at the speed of 1500 km $h^{-1}$ relative to the jet plane. What is the speed of the latter with respect to an observer on ground?
24. What do you understand by a non-uniform motion? Explain, average velocity and instantaneous velocity of an object moving along a straight line.

## SECTION-C

### Short Answer (SA) Type Questions

25. (i) What do you mean by free fall?
    (ii) Taking upwards direction positive, point of projection as origin and neglecting air resistance, draw the position-time, velocity-time and acceleration-time graphs of an object under free fall.
26. A boy travelling along a straight line, traversed one third of the total distance with a velocity of 4 m/s. The remaining part of the distance was covered with a velocity of 2 m/s and 6 m/s for the equal time intervals. Calculate the average velocity of the boy during his entire journey.
27. A ball is dropped and its displacement vs time graph is as shown figure. (displacement $x$ is from ground and all quantities are +ve upwards).




(a) Plot qualitatively velocity vs time graph.
(b) Plot qualitatively acceleration vs time graph.

28. A woman starts from her home at 9.00 am, walks with a speed of 5 km $h^{-1}$ on a straight road up to her office 2.5 km away, stays at the office up to 5.00 pm, and returns home by an auto with a speed of 25 km $h^{-1}$. Choose suitable scales and plot the $x$-$t$ graph of her motion.

## SECTION-D

### Long Answer (LA) Type Questions

29. What do the following represent? [$x$ = position, $a$ = acceleration, $v$ = velocity, $s$ = distance]
    (i) Slope of $x$-$t$ graph
    (ii) Slope of $v$-$t$ graph
    (iii) Area of $v$-$t$ graph
    (iv) Area of $a$-$t$ graph
    (v) Slope of $s$-$t$ graph
30. A motor car moving at a speed of 72 km/h cannot come to a stop in less than 3.0 s while for a truck this time interval is 5.0 s. On a highway the car is behind the truck both moving at 72 km/h. The truck gives a signal that it is going to stop at emergency. At what distance the car should be from the truck so that it does not bump onto (collide with) the truck? Human response time is 0.5 s.
31. A police van moving on a highway with a speed of 30 km $h^{-1}$ fires a bullet at a thief's car speeding away in the same direction with a speed of 192 km $h^{-1}$. If the muzzle speed of the bullet is 150 m $s^{-1}$, with what speed does the bullet hit the thief's car? (Note: Obtain that speed which is relevant for damaging the thief's car).

## SECTION-E

### Case Study Based Questions

The velocity of an object depends on the frame of reference. For example a man can run on ground with a speed of say 8 m/s. This means that the velocity of person w.r.t. ground is 8 m/s. Now suppose the same person stands on the roof of a train moving with a speed of 10 m/s. If man does not walk on the train, he will still move with respect to ground with a speed of 10 m/s. Here we will say that velocity of man w.r.t. train is zero while the velocity of man w.r.t. ground is 10 m/s. Now if the man begins to run on the roof with speed 8 m/s in forward direction his velocity w.r.t. train will be 8 m/s and w.r.t. ground it will be 18 m/s. If man runs backward its velocity w.r.t. ground will be 2 m/s.

In general

$$\vec{v}_{M/G} = \vec{v}_{M/T} + \vec{v}_{T/G}$$

Here, $\vec{v}_{M/G}$ is velocity of man w.r.t. ground

$\vec{v}_{M/T}$ is velocity of man w.r.t. train

$\vec{v}_{T/G}$ is velocity of train w.r.t. ground

Similarly a boat can be rowed in still water at a speed $u$. The speed of boat w.r.t. ground, while moving in a stream of water flowing at speed $v$, will be $u + v$ while moving downstream (forward) and $u - v$ while moving upstream (backward). Based on the above information, answer the following questions.

32. A person can swim in still water at a speed 5 m/s. How much time will he take to move a distance of 120 m in stream of water flowing at 1 m/s upstream and downstream respectively?

(1) 24 s, 24 s (2) 20 s, 30 s

(3) 30 s, 20 s (4) 30 s, 24 s

33. A person covers a certain distance downstream in 15 s and upstream in 30 s. How much time will be take to cover the same distance in still water?

(1) 22.5 s (2) $15\sqrt{2}$ s

(3) 20 s (4) None of these

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